

Predicting Heart Disease

Can we tell which patients have heart disease?

The Goal

Predict whether a patient has heart disease (Yes or No)

Why It Matters

Catching it early can save lives

The Risk

Two kinds of mistakes can happen

False Positive

A healthy patient is flagged as sick and gets extra tests

False Negative (worse)

A sick patient is missed and gets no treatment

Ameer Adegun, DATA 201 Project 1

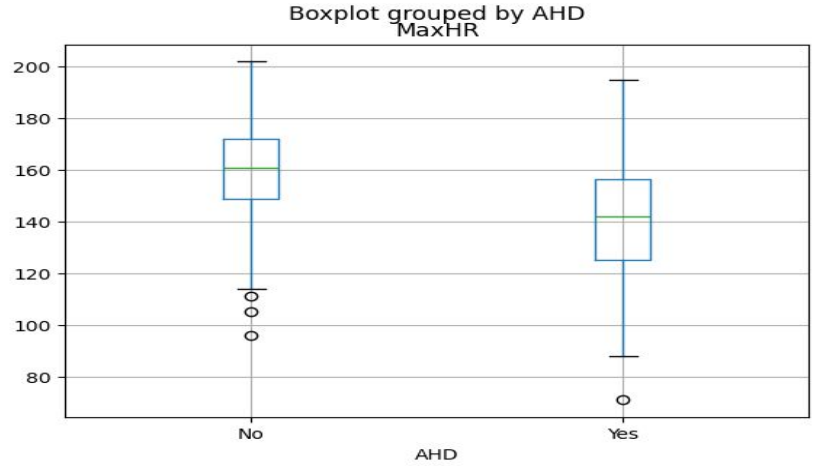
The Heart Disease Dataset

What we are working with

297
patients after removing rows with missing values

13
features, like age, chest pain type, and max heart rate

164 vs 139
patients without vs with heart disease, fairly balanced

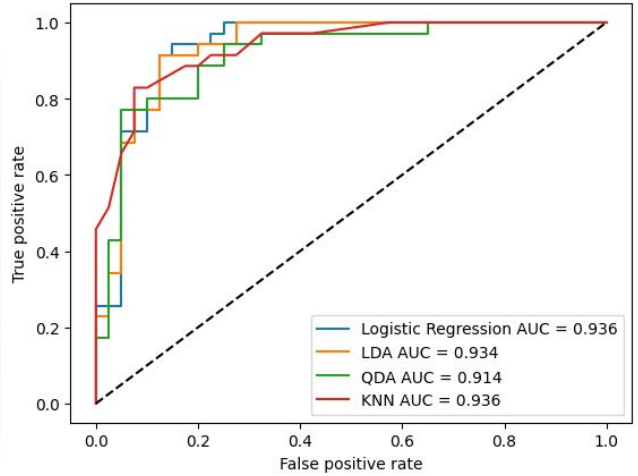


Patients with heart disease tend to have a lower max heart rate

Model Comparison

Four models tested at a 0.40 threshold

Model	Accuracy	Recall	Specificity	AUC
Logistic Regression	0.880	0.914	0.850	0.936
LDA	0.880	0.886	0.875	0.934
QDA	0.813	0.829	0.800	0.914
KNN (K=16)	0.840	0.886	0.800	0.936



Key result: Logistic Regression has the best recall (0.914) and ties for the best AUC (0.936)



Which Model I Would Use

My decision: Logistic Regression

Recall (0.914): only 3 sick patients missed

False Negative
A sick patient goes untreated, the worst error

False Positive
A healthy patient gets extra tests, less harmful

Limitation
Only 297 patients, so it needs testing on more hospitals